

MSE-1 Qusetion Paper and Answer LLM & Prompt Engineering

Q1: What is the purpose of using a system message in a prompt?

Answer: To define the behavior or role of the AI model at the start of a conversation.

Q2: Identify the type of prompting used in the following instruction:

"Translate the sentence into Hindi: 'How are you?'"

Answer: Zero-shot prompting

Q3: What does the temperature parameter control in a language model?

Answer: It controls the randomness or creativity of the model's output.

Q4: Identify the prompting pattern used in the following instruction:

"Let's solve this step-by-step."

Answer: Chain-of-thought prompting

Q5: Explain the role of context in a prompt with a simple example.

Answer:

Context provides background information to help the model respond more accurately.

Example:

Instruction: *Summarize the following legal paragraph.*

Context: *This is part of a contract law agreement.*

Q6: Construct a one-shot prompt to convert temperatures from Celsius to Fahrenheit.

Answer:

Example:

"Convert Celsius to Fahrenheit.

Example: 0°C → 32°F

Now convert: 25°C → ""*

Q7: Explain the role of the **top-p** parameter in prompt engineering.

Answer:

Top-p controls nucleus sampling by choosing from the smallest set of top tokens whose cumulative probability is greater than or equal to p . It ensures only the most likely tokens are sampled, making the output more focused.

Q8: Construct a prompt to generate a short story using a specific **persona** of a humorous children's book author.

Answer:

Prompt: "You are a humorous children's book author. Write a short story about a talking cat that becomes the mayor of a small town. Use playful language and make it suitable for 7–10-year-olds."

Q9: Analyze the following prompt and identify the missing components. Then, rewrite it as a complete and effective prompt using proper structure.

Prompt Given:

"Translate this into Spanish: 'Good Morning.'"

Answer:

- The prompt is **simple** but **lacks context** and **explicit instruction structure**.
- It does not specify **who** the user is, the **output format**, or whether the response should include explanation, translation only, or tone (formal/informal).

Rewritten:

You are a language translation assistant. Your task is to accurately translate simple English sentences into Spanish.

Instruction: Translate the following English sentence into Spanish.

Context: This is for a travel guide aimed at tourists.

Input: "Good Morning"

Output: Provide only the translated sentence.

Q10: You are developing a customer support chatbot using an LLM. Apply appropriate advanced prompt engineering techniques (such as persona creation, temperature tuning, and prompt design patterns) to ensure that the chatbot provides helpful, friendly, and consistent responses. Illustrate with an example prompt.

Answer:

Application of Techniques:

- **Persona Creation:** Define the chatbot's role clearly (e.g., "You are a polite and friendly customer support assistant.")
- **Temperature Tuning:** Set temperature low (e.g., 0.3–0.5) for reliable and factual responses.
- **Prompt Design Pattern:** Use Chain-of-Thought or Self-Ask to break down complex customer queries.

Example Prompt:

System Prompt:

"You are a helpful and friendly customer support assistant for an electronics company. Always greet the customer politely, answer clearly, and offer step-by-step instructions when needed."

User Prompt:

"My new headphones are not connecting to Bluetooth. What should I do?"

Expected LLM Behavior:

Greets user → Clarifies issue → Provides 2–3 step solution in polite tone.

Q11: Analyze the concept of prompt design patterns used in advanced prompt engineering. Choose any two patterns and explain, with examples, how each pattern improves the reasoning or response quality of a language model.

Answer:

Prompt design patterns are structured strategies for crafting prompts that guide a language model's reasoning, task handling, or interaction process. By analyzing how the model processes tasks under different patterns, we can better understand and improve its performance.

1. Self-Ask Pattern

➤ **Structure:** Breaks complex queries into sub-questions.

➤ **Example:**

Question: "What is the capital of the country where the Great Wall is located?"

Model Process:

- Q1: Where is the Great Wall located? → China
- Q2: What is the capital of China? → Beijing
- **Final Answer:** Beijing

- **Improvement:** Helps the model handle multi-hop reasoning by sequentially answering simpler questions, reducing the chance of logical skips.

2. ReAct (Reasoning + Acting) Pattern

- **Structure:** Alternates between internal thoughts and external actions like tool use or memory retrieval.
- **Example:**

Question: What is the current population of Brazil?

Thought: I need up-to-date data.

Action: Search[Population of Brazil]

Observation: 214 million

Final Answer: Brazil's population is approximately 214 million.

- **Improvement:** Combines step-by-step logic with real-time data retrieval, making responses more accurate, explainable, and grounded in facts.

By analyzing the function of these patterns, we see that they enhance the **model's interpretability, reasoning clarity, and reliability**, especially in complex tasks.